Programming Code for Arduino Uno Software

```
#include <Wire.h>                    //main code for including the Wire library in the program

#include <LiquidCrystal_I2C.h>       //main code for including the I2C LCD library in the program

int sensorPin0 = A0;          // creates a new sensorPin0 variable using analog pin 0 as the input
int sensorPin1 = A1;          // creates a new sensorPin1 variable using analog pin 1 as the input
int sensorPin2 = A2;          // creates a new sensorPin2 variable using analog pin 2 as the input
int sensorPin3 = A3;          // creates a new sensorPin3 variable using analog pin 3 as the input

float sensorValue0 = 0;       // creates a floating-point variable for the value from sensor 0
float sensorValue1 = 0;       // creates a floating-point variable for the value from sensor 1
float sensorValue2 = 0;       // creates a floating-point variable for the value from sensor 2
float sensorValue3 = 0;       // creates a floating-point variable for the value from sensor 3

float x1;                     // variable for potentiometer 1
float x2;                     // variable for potentiometer 2
float x3;                     //variable for potentiometer 3
float x4;                     // variable for potentiometer 4
float multimeter;             // variable for the multimeter value

float Y1;

float Y2;

float Y3;

float Y4;

float Y5;

float Y6;

float Y7;
```

```
float Y8;
float Y9;
float Y10;
float Y11;
float Y12;
float Y13;
float Y14;
float Y15;
float Y16;
float Y17;
float Y18;
float Y19;
LiquidCrystal_I2C lcd1(0x27,16,2);          // initializes an I2C LCD; the original comment
                                            specifies a 4-row, 20-column display
LiquidCrystal_I2C lcd2(0x26,16,2);          // display on LCD 2
void setup() {          // the setup() function initializes the sketch and runs only once each
                        time the Arduino board restarts.
lcd1.init();            // initializes the LCD
lcd1.backlight();       // turns on the backlight
lcd1.setCursor(0,0);
lcd1.print ("Soil Moisture");          // lcd 1
lcd2.init();                           // initializes the LCD
lcd2.backlight();               // turns on the backlight
lcd2.setCursor(0,0);
lcd2.print ("Soil Moisture");          // lcd 2
```

```
delay(1000);

lcd1.clear();

lcd2.clear();


Serial.begin(9600);                // enables the Serial Monitor at 9600 baud
}
void loop(){              //the loop() function repeatedly executes the program commands to
                          control the Arduino board, including reading inputs and changing
                          outputs

sensorValue0 = analogRead(sensorPin0);     // reading from potentiometer 1

sensorValue1 = analogRead(sensorPin1);     // reading from potentiometer 2

sensorValue2 = analogRead(sensorPin2);     // reading from potentiometer 3

sensorValue3 = analogRead(sensorPin3);     // reading from potentiometer 4

x1=((0.097*sensorValue0)-0.039);    // resistance value for potentiometer 1

x2=((0.097*sensorValue1)-0.039);    // resistance value for potentiometer 2

x3=((0.097*sensorValue2)-0.039);    // resistance value for potentiometer 3

x4=((0.097*sensorValue3)-0.039);    // resistance value for potentiometer 4

multimeter=(x3+x4);                 // resistance value calculated for the multimeter


Serial.print("P1=");

Serial.println(x1);

delay (1000);

Serial.print("P2=");

Serial.println(x2);

delay(1000);
```

```
Serial.print("P3=");

Serial.println(x3);

delay(1000);

Serial.print("P4=");

Serial.println(x4);

delay(1000);

Serial.print("MM=");

Serial.println(multimeter);

delay(1000);

Serial.print("KAT");

Serial.println(Y19);

delay (500);

Serial.print(" " );

lcd1.setCursor(0,0);

lcd1.print ("A:");

//displays the value of potentiometer 1 on the LCD

//delay(1000);

lcd1.setCursor(3,0);

lcd1.print (x1);

lcd1.setCursor(9,0);


lcd1.print (" ");

delay(10);
```

```
lcd1.setCursor(0,1);
lcd1.print ("B:");
//displays the value of potentiometer 2 on the LCD
//delay(1000);
lcd1.setCursor(3,1);
lcd1.print (x2 );
lcd1.setCursor(9,1);

lcd1.print (" ");
//delay(1000);
delay(10);
 lcd1.setCursor(10,0);

lcd1.print ("C:");
//delay(1000);
lcd1.setCursor(12,0);
//displays the value of potentiometer 3 on the LCD
lcd1.print (x3);
delay(10);
lcd1.setCursor(10,1);
lcd1.print ("D:");
//displays the value of potentiometer 4 on the LCD
//delay(1000);
lcd1.setCursor(12,1);
```

```
lcd1.print (x4);
//delay(1000);
delay(10);

lcd2.setCursor(0,0);
lcd2.print ("R=");
//delay(1000);
lcd2.setCursor(3,0);
lcd2.print (multimeter);
lcd2.setCursor(9,0);
lcd2.print (" ");
delay(10);

if (x1 > 0 && x1< 32){
lcd2.setCursor(10,0);

//displays the soil-type code on the LCD
lcd2.print("JT:");
lcd2.setCursor(14,0);
lcd2.print("a");
delay(10);
}

if (x1> 33 && x1 < 66){
```

```
lcd2.setCursor(10,0);

//displays the soil-type code on the LCD

lcd2.print("JT:");

lcd2.setCursor(14,0);

lcd2.print("b");

delay(10);

}

if (x1 > 66 && x1 <100) {

 lcd2.setCursor(10,0);
lcd2.print("JT:");

//displays the soil-type code on the LCD

lcd2.setCursor(14,0);

lcd2.print("c");

delay(10);

}

// condition for the soil-temperature parameter

if (x2 > 0 && x2 < 32){

Serial.println (" temperature: 24 degrees ");

lcd2.setCursor(0,1);

//displays the temperature parameter on the LCD
```

```
lcd2.print("T= 24 ");

delay(10);

}

if (x2 > 33 && x2 < 66 ){

Serial.println (" temperature: 28 degrees ");

lcd2.setCursor(0,1);

//displays the temperature parameter on the LCD

lcd2.print("T = 28");

delay(10);

}

if (x2 > 67 && x2 < 100){

Serial.println (" temperature: 32 degrees ");

lcd2.setCursor(0,1);

//displays the temperature parameter on the LCD

lcd2.print("T = 32");

delay(10);

}


if ((x1 > 0 && x1< 32)&& (x2 > 0 && x2 < 32)){

 Y1 = (((10.374678*1)/3) - ((6.2961398*24)/32) - ((6.6195249*multimeter)/22) -
(0.78652129));

Y2 = (((-5.2774464*1)/3) - ((8.6484264*24)/32) - ((9.6981703*multimeter)/22) +
(14.854810));
```

```
Y3 = (((3.8060959*1)/3) + ((27.236429*24)/32) + ((6.4649224*multimeter)/22) -
(25.437087));

Y4 = 1/((1+exp(-Y1)) );

Y5 = 1/((1+exp(-Y2)) );

Y6 = 1/((1+exp(-Y3)) );

Y7 = ((-2.8717559*Y4) - (6.0005746*Y5) - (11.496338*Y6) + (11.630656));

Y8 = ((10.811394*Y4) - (1.6096994*Y5) - (0.19157851*Y6) - (0.85376095));

Y9 = ((-9.9052177*Y4) + (4.1450168*Y5) + (3.8937926*Y6) - (5.0756358));

Y10 = ((2.3690141*Y4) + (1.5123419*Y5) + (0.11309993*Y6) - (6.7704054));

Y11 = ((10.974770*Y4) - (5.9954970*Y5) - (8.6160664*Y6) + (7.5410023));

Y12 = ((1-exp(-2*Y7)))/((1+exp(-2*Y7)) );

Y13 = ((1-exp(-2*Y8)))/((1+exp(-2*Y8)) );

Y14 = ((1-exp(-2*Y9)))/((1+exp(-2*Y9)) );

Y15 = ((1-exp(-2*Y10)))/((1+exp(-2*Y10)) );

Y16 = ((1-exp(-2*Y11)))/((1+exp(-2*Y11)) );

Y17 = ((-3.3195649*Y12) - (8.7607633*Y13) + (3.1435480*Y14) - (0.79172933*Y15) +
(7.4413874*Y16) + (0.65732827));

Y18 = 1/((1+exp(-Y17)) );

Y19 = (Y18*(65.9));

}

else if ((x1 > 0 && x1< 32)&& (x2 > 33 && x2 < 66)){
```

Y1 = (((10.374678*1)/3) - ((6.2961398*28)/32) - ((6.6195249*multimeter)/22) - (0.78652129));

Y2 = (((-5.2774464*1)/3) - ((8.6484264*28)/32) - ((9.6981703*multimeter)/22) + (14.854810));

Y3 = (((3.8060959*1)/3) + ((27.236429*28)/32) + ((6.4649224*multimeter)/22) - (25.437087));

Y4 = 1/((1+exp(-Y1)));

Y5 = 1/((1+exp(-Y2)));

Y6 = 1/((1+exp(-Y3)));

Y7 = ((-2.8717559*Y4) - (6.0005746*Y5) - (11.496338*Y6) + (11.630656));

Y8 = ((10.811394*Y4) - (1.6096994*Y5) - (0.19157851*Y6) - (0.85376095));

Y9 = ((-9.9052177*Y4) + (4.1450168*Y5) + (3.8937926*Y6) - (5.0756358));

Y10 = ((2.3690141*Y4) + (1.5123419*Y5) + (0.11309993*Y6) - (6.7704054));

Y11 = ((10.974770*Y4) - (5.9954970*Y5) - (8.6160664*Y6) + (7.5410023));

Y12 = ((1-exp(-2*Y7)))/((1+exp(-2*Y7)));

Y13 = ((1-exp(-2*Y8)))/((1+exp(-2*Y8)));

Y14 = ((1-exp(-2*Y9)))/((1+exp(-2*Y9)));

Y15 = ((1-exp(-2*Y10)))/((1+exp(-2*Y10)));

Y16 = ((1-exp(-2*Y11)))/((1+exp(-2*Y11)));

Y17 = ((-3.3195649*Y12) - (8.7607633*Y13) + (3.1435480*Y14) - (0.79172933*Y15) + (7.4413874*Y16) + (0.65732827));

Y18 = 1/((1+exp(-Y17)));

Y19 = (Y18*(65.9));

```
}

else if ((x1 > 0 && x1< 32)&& (x2 > 67 && x2 < 100)){

  Y1 = (((10.374678*1)/3) - ((6.2961398*32)/32) - ((6.6195249*multimeter)/22) -
(0.78652129));

Y2 = (((-5.2774464*1)/3) - ((8.6484264*32)/32) - ((9.6981703*multimeter)/22) +
(14.854810));

Y3 = (((3.8060959*1)/3) + ((27.236429*32)/32) + ((6.4649224*multimeter)/22) -
(25.437087));

Y4 = 1/((1+exp(-Y1)) );

Y5 = 1/((1+exp(-Y2)) );

Y6 = 1/((1+exp(-Y3)) );

Y7 = ((-2.8717559*Y4) - (6.0005746*Y5) - (11.496338*Y6) + (11.630656));

Y8 = ((10.811394*Y4) - (1.6096994*Y5) - (0.19157851*Y6) - (0.85376095));

Y9 = ((-9.9052177*Y4) + (4.1450168*Y5) + (3.8937926*Y6) - (5.0756358));

Y10 = ((2.3690141*Y4) + (1.5123419*Y5) + (0.11309993*Y6) - (6.7704054));

Y11 = ((10.974770*Y4) - (5.9954970*Y5) - (8.6160664*Y6) + (7.5410023));

Y12 = ((1-exp(-2*Y7)))/((1+exp(-2*Y7)) );

Y13 = ((1-exp(-2*Y8)))/((1+exp(-2*Y8)) );

Y14 = ((1-exp(-2*Y9)))/((1+exp(-2*Y9)) );

Y15 = ((1-exp(-2*Y10)))/((1+exp(-2*Y10)) );

Y16 = ((1-exp(-2*Y11)))/((1+exp(-2*Y11)) );
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Y17 = ((-3.3195649*Y12) - (8.7607633*Y13) + (3.1435480*Y14) - (0.79172933*Y15) + (7.4413874*Y16) + (0.65732827));

Y18 = 1/((1+exp(-Y17)));

Y19 = (Y18*(65.9)); }

else if ((x1 >33 && x1< 66)&& (x2 > 0 && x2 < 32)){

Y1 = (((10.374678*2)/3) - ((6.2961398*24)/32) - ((6.6195249*multimeter)/22) - (0.78652129));

Y2 = (((-5.2774464*2)/3) - ((8.6484264*24)/32) - ((9.6981703*multimeter)/22) + (14.854810));

Y3 = (((3.8060959*2)/3) + ((27.236429*24)/32) + ((6.4649224*multimeter)/22) - (25.437087));

Y4 = 1/((1+exp(-Y1)));

Y5 = 1/((1+exp(-Y2)));

Y6 = 1/((1+exp(-Y3)));

Y7 = ((-2.8717559*Y4) - (6.0005746*Y5) - (11.496338*Y6) + (11.630656));

Y8 = ((10.811394*Y4) - (1.6096994*Y5) - (0.19157851*Y6) - (0.85376095));

Y9 = ((-9.9052177*Y4) + (4.1450168*Y5) + (3.8937926*Y6) - (5.0756358));

Y10 = ((2.3690141*Y4) + (1.5123419*Y5) + (0.11309993*Y6) - (6.7704054));

Y11 = ((10.974770*Y4) - (5.9954970*Y5) - (8.6160664*Y6) + (7.5410023));

Y12 = ((1-exp(-2*Y7)))/((1+exp(-2*Y7)));

Y13 = ((1-exp(-2*Y8)))/((1+exp(-2*Y8)));

Y14 = ((1-exp(-2*Y9)))/((1+exp(-2*Y9)));

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Y15 = ((1-exp(-2*Y10)))/((1+exp(-2*Y10)) );

Y16 = ((1-exp(-2*Y11)))/((1+exp(-2*Y11)) );

Y17 = ((-3.3195649*Y12) - (8.7607633*Y13) + (3.1435480*Y14) - (0.79172933*Y15) +
(7.4413874*Y16) + (0.65732827));

Y18 = 1/((1+exp(-Y17)) );

Y19 = (Y18*(65.9));

}

else if ((x1 > 33 && x1< 66)&& (x2 > 33 && x2 < 66)){

 Y1 = (((10.374678*2)/3) - ((6.2961398*28)/32) - ((6.6195249*multimeter)/22) -
(0.78652129));

Y2 = (((-5.2774464*2)/3) - ((8.6484264*28)/32) - ((9.6981703*multimeter)/22) +
(14.854810));

Y3 = (((3.8060959*2)/3) + ((27.236429*28)/32) + ((6.4649224*multimeter)/22) -
(25.437087));

Y4 = 1/((1+exp(-Y1)) );

Y5 = 1/((1+exp(-Y2)) );

Y6 = 1/((1+exp(-Y3)) );

Y7 = ((-2.8717559*Y4) - (6.0005746*Y5) - (11.496338*Y6) + (11.630656));

Y8 = ((10.811394*Y4) - (1.6096994*Y5) - (0.19157851*Y6) - (0.85376095));

Y9 = ((-9.9052177*Y4) + (4.1450168*Y5) + (3.8937926*Y6) - (5.0756358));

Y10 = ((2.3690141*Y4) + (1.5123419*Y5) + (0.11309993*Y6) - (6.7704054));

Y11 = ((10.974770*Y4) - (5.9954970*Y5) - (8.6160664*Y6) + (7.5410023));

Y12 = ((1-exp(-2*Y7)))/((1+exp(-2*Y7)) );
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Y13 = ((1-exp(-2*Y8)))/((1+exp(-2*Y8)) );

Y14 = ((1-exp(-2*Y9)))/((1+exp(-2*Y9)) );

Y15 = ((1-exp(-2*Y10)))/((1+exp(-2*Y10)) );

Y16 = ((1-exp(-2*Y11)))/((1+exp(-2*Y11)) );

Y17 = ((-3.3195649*Y12) - (8.7607633*Y13) + (3.1435480*Y14) - (0.79172933*Y15) +
(7.4413874*Y16) + (0.65732827));

Y18 = 1/((1+exp(-Y17)) );

Y19 = (Y18*(65.9));

}


else if ((x1 > 33 && x1< 66) && (x2 > 67 && x2 < 100)){

 Y1 = (((10.374678*2)/3) - ((6.2961398*32)/32) - ((6.6195249*multimeter)/22) -
(0.78652129));

Y2 = (((-5.2774464*2)/3) - ((8.6484264*32)/32) - ((9.6981703*multimeter)/22) +
(14.854810));

Y3 = (((3.8060959*2)/3) + ((27.236429*32)/32) + ((6.4649224*multimeter)/22) -
(25.437087));


Y4 = 1/((1+exp(-Y1)) );

Y5 = 1/((1+exp(-Y2)) );

Y6 = 1/((1+exp(-Y3)) );


Y7 = ((-2.8717559*Y4) - (6.0005746*Y5) - (11.496338*Y6) + (11.630656));

Y8 = ((10.811394*Y4) - (1.6096994*Y5) - (0.19157851*Y6) - (0.85376095));

Y9 = ((-9.9052177*Y4) + (4.1450168*Y5) + (3.8937926*Y6) - (5.0756358));

Y10 = ((2.3690141*Y4) + (1.5123419*Y5) + (0.11309993*Y6) - (6.7704054));
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Y11 = ((10.974770*Y4) - (5.9954970*Y5) - (8.6160664*Y6) + (7.5410023));

Y12 = ((1-exp(-2*Y7)))/((1+exp(-2*Y7)) );

Y13 = ((1-exp(-2*Y8)))/((1+exp(-2*Y8)) );

Y14 = ((1-exp(-2*Y9)))/((1+exp(-2*Y9)) );

Y15 = ((1-exp(-2*Y10)))/((1+exp(-2*Y10)) );

Y16 = ((1-exp(-2*Y11)))/((1+exp(-2*Y11)) );

Y17 = ((-3.3195649*Y12) - (8.7607633*Y13) + (3.1435480*Y14) - (0.79172933*Y15) +
(7.4413874*Y16) + (0.65732827));

Y18 = 1/((1+exp(-Y17)) );

Y19 = (Y18*(65.9));

}

else if ((x1 >67 && x1< 100) && (x2 > 0  &&  x2 < 32)){

 Y1 = (((10.374678*3)/3) - ((6.2961398*24)/32) - ((6.6195249*multimeter)/22) -
(0.78652129));

Y2 = (((-5.2774464*3)/3) - ((8.6484264*24)/32) - ((9.6981703*multimeter)/22) +
(14.854810));

Y3 = (((3.8060959*3)/3) + ((27.236429*24)/32) + ((6.4649224*multimeter)/22) -
(25.437087));

Y4 = 1/((1+exp(-Y1)) );

Y5 = 1/((1+exp(-Y2)) );

Y6 = 1/((1+exp(-Y3)) );

Y7 = ((-2.8717559*Y4) - (6.0005746*Y5) - (11.496338*Y6) + (11.630656));
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```
Y8 = ((10.811394*Y4) - (1.6096994*Y5) - (0.19157851*Y6) - (0.85376095));

Y9 = ((-9.9052177*Y4) + (4.1450168*Y5) + (3.8937926*Y6) - (5.0756358));

Y10 = ((2.3690141*Y4) + (1.5123419*Y5) + (0.11309993*Y6) - (6.7704054));

Y11 = ((10.974770*Y4) - (5.9954970*Y5) - (8.6160664*Y6) + (7.5410023));


Y12 = ((1-exp(-2*Y7)))/((1+exp(-2*Y7)) );

Y13 = ((1-exp(-2*Y8)))/((1+exp(-2*Y8)) );

Y14 = ((1-exp(-2*Y9)))/((1+exp(-2*Y9)) );

Y15 = ((1-exp(-2*Y10)))/((1+exp(-2*Y10)) );

Y16 = ((1-exp(-2*Y11)))/((1+exp(-2*Y11)) );

Y17 = ((-3.3195649*Y12) - (8.7607633*Y13) + (3.1435480*Y14) - (0.79172933*Y15) +
(7.4413874*Y16) + (0.65732827));

Y18 = 1/((1+exp(-Y17)) );

Y19 = (Y18*(65.9));

}


else if ((x1 > 67 && x1< 100) && (x2 > 33 && x2 < 66)){

  Y1 = (((10.374678*3)/3) - ((6.2961398*28)/32) - ((6.6195249*multimeter)/22) -
(0.78652129));

Y2 = (((-5.2774464*3)/3) - ((8.6484264*28)/32) - ((9.6981703*multimeter)/22) +
(14.854810));

Y3 = (((3.8060959*3)/3) + ((27.236429*28)/32) + ((6.4649224*multimeter)/22) -
(25.437087));


Y4 = 1/((1+exp(-Y1)) );

Y5 = 1/((1+exp(-Y2)) );
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```
Y6 = 1/((1+exp(-Y3)) );

Y7 = ((-2.8717559*Y4) - (6.0005746*Y5) - (11.496338*Y6) + (11.630656));

Y8 = ((10.811394*Y4) - (1.6096994*Y5) - (0.19157851*Y6) - (0.85376095));

Y9 = ((-9.9052177*Y4) + (4.1450168*Y5) + (3.8937926*Y6) - (5.0756358));

Y10 = ((2.3690141*Y4) + (1.5123419*Y5) + (0.11309993*Y6) - (6.7704054));

Y11 = ((10.974770*Y4) - (5.9954970*Y5) - (8.6160664*Y6) + (7.5410023));

Y12 = ((1-exp(-2*Y7)))/((1+exp(-2*Y7)) );

Y13 = ((1-exp(-2*Y8)))/((1+exp(-2*Y8)) );

Y14 = ((1-exp(-2*Y9)))/((1+exp(-2*Y9)) );

Y15 = ((1-exp(-2*Y10)))/((1+exp(-2*Y10)) );

Y16 = ((1-exp(-2*Y11)))/((1+exp(-2*Y11)) );

Y17 = ((-3.3195649*Y12) - (8.7607633*Y13) + (3.1435480*Y14) - (0.79172933*Y15) +
(7.4413874*Y16) + (0.65732827));

Y18 = 1/((1+exp(-Y17)) );

Y19 = (Y18*(65.9));}

else if ((x1 >67 && x1< 100)&& (x2 > 67 && x2 < 100)){

  Y1 = (((10.374678*3)/3) - ((6.2961398*32)/32) - ((6.6195249*multimeter)/22) -
(0.78652129));

Y2 = (((-5.2774464*3)/3) - ((8.6484264*32)/32) - ((9.6981703*multimeter)/22) +
(14.854810));

Y3 = (((3.8060959*3)/3) + ((27.236429*32)/32) + ((6.4649224*multimeter)/22) -
(25.437087));
```

```
Y4 = 1/((1+exp(-Y1)) );

Y5 = 1/((1+exp(-Y2)) );

Y6 = 1/((1+exp(-Y3)) );

Y7 = ((-2.8717559*Y4) - (6.0005746*Y5) - (11.496338*Y6) + (11.630656));

Y8 = ((10.811394*Y4) - (1.6096994*Y5) - (0.19157851*Y6) - (0.85376095));

Y9 = ((-9.9052177*Y4) + (4.1450168*Y5) + (3.8937926*Y6) - (5.0756358));

Y10 = ((2.3690141*Y4) + (1.5123419*Y5) + (0.11309993*Y6) - (6.7704054));

11 = ((10.974770*Y4) - (5.9954970*Y5) - (8.6160664*Y6) + (7.5410023));

Y12 = ((1-exp(-2*Y7)))/((1+exp(-2*Y7)) );

Y13 = ((1-exp(-2*Y8)))/((1+exp(-2*Y8)) );

Y14 = ((1-exp(-2*Y9)))/((1+exp(-2*Y9)) );

Y15 = ((1-exp(-2*Y10)))/((1+exp(-2*Y10)) );

Y16 = ((1-exp(-2*Y11)))/((1+exp(-2*Y11)) );

Y17 = ((-3.3195649*Y12) - (8.7607633*Y13) + (3.1435480*Y14) - (0.79172933*Y15) +
(7.4413874*Y16) + (0.65732827));

Y18 = 1/((1+exp(-Y17)) );

Y19 = (Y18*(65.9));

}

lcd2.setCursor(7,1);

lcd2.print ("KAT=");

//displays the final soil-moisture value on the LCD
```

```
lcd2.setCursor(11,1);

lcd2.print(Y19);

delay(10);

}
```